Q/R/M/C: A Stage-Decomposition Evaluation Protocol for Memory-Based Agents

Anonymous submission

Abstract

Success-only metrics collapse qualitatively different memory failures into a single scalar and obscure which stage of the retrieval pipeline fails. We describe an evaluation protocol, the Q/R/M/C decomposition (Query formation, Retrieval satisfaction, target Materialization, Completion), operationalised so that four conditional rates can be computed from standard trajectory logs. The arithmetic is an elementary chain rule over logged events; the contribution lies in the event definitions, which make the rates estimable in practice, analogously to precision and recall, whose value lies in their definitions rather than in their arithmetic. In single-agent experiments (MiniGrid, BabyAI), oracle-stage interventions separate a retrieval-limited task ($0.39 \rightarrow 0.60$ under oracle retrieval) from downstream-limited ones, and log inspection refines the diagnosis: ranking among retrieved candidates is near-optimal (96.9%), but the candidate set is empty at 91% of decision points, so the limiting factor is memory accumulation. In multi-agent experiments (35,640 runs, four sharing architectures, cadence K), the predicted bottleneck shift between Materialization and Completion is confirmed as cluster-robust slopes (Spearman $-0.53/+0.43$), time-to-success among successes has an interior optimum at $K{=}8$, and a provenance annotation identifies the mechanism: under fast sharing the completion collapse is concentrated in locks backed by peers' evidence. The same tool-trace contract also instruments three third-party agent stacks (OpenAI SDK, LangGraph, AutoGen). Engineered structural failures localise identically on every stack ($R^{\star}{=}0$, $C^{\star}{=}0$ in $20/20$), while on clean tasks the frameworks spread 0.50–0.75 end-to-end, and the stage profiles attribute the spread to low first-commitment accuracy (M_1) combined with framework-specific recovery behaviour. The protocol makes memory failures comparable across symbolic and LLM systems, single- and multi-agent, wherever a query/lock interface is exposed; on opaque learned policies it measures whether the stages are observable at all.

Introduction

Embodied, goal-conditioned, and LLM agents increasingly retrieve places, objects, and facts *by meaning* from an explicit memory, yet they are evaluated almost exclusively end-to-end: if success improves, memory is judged useful. Two systems with the same success rate can fail for different reasons: one may fail to accumulate queryable evidence, while another retrieves correctly but loses the target to a peer agent. This paper describes a measurement instrument that separates such cases.

The instrument is a four-stage decomposition of memory-based task completion: **Q**uery formation (the planner emits a semantic intent), **R**etrieval satisfaction (memory returns a candidate matching the task), target **M**aterialization (the planner locks one candidate as an actionable target), and **C**ompletion (the controller reaches it). Logged as per-episode events $Q^{\star}, R^{\star}, M^{\star}, C^{\star}$, they factorise semantic-path completion into conditional rates estimable from ordinary trajectory logs (Definitions 1–2, §). The consistency argument is the chain rule plus frequency estimation, and we claim no mathematical novelty for it. The precedent is the decomposition of accuracy into precision and recall, which was also arithmetically trivial and still changed evaluation practice. Our contribution is a set of log-level event definitions and the diagnoses they support.

Related Work

Cognitive-map and semantic-map lines motivate explicit place abstraction (Tolman 1948; Gupta et al. 2017; Savinov, Dosovitskiy, and Koltun 2018); goal-conditioned RL conditions on goals or embeddings (Andrychowicz et al. 2017); multi-agent RL studies learned channels (Sukhbaatar, Szlam, and Fergus 2016; Foerster et al. 2018) and gossip protocols (Boyd et al. 2006); LLM-agent systems add explicit memory layers evaluated end-to-end (Wang et al. 2024; Shinn et al. 2023; Packer et al. 2023; Park et al. 2023; Wu et al. 2023). The protocol is orthogonal to these lines: it proposes no architecture, makes any of them stage-diagnosable from logs, and yields a slope-level cadence prediction testable on those logs.

Stage-wise evaluation. The closest line decomposes retrieval-augmented generation: RAGAS scores retrieval and generation quality separately (Es et al. 2024), RAGChecker refines this to claim-level diagnostics meta-evaluated against human judgments (Ru et al. 2024), and AgentBoard replaces final success with analytic progress rates for multi-turn LLM agents (Ma et al. 2024). Q/R/M/C differs on three axes: its events are logged actions of a task-executing agent, not LLM-judged text quality; it models commitment (M) and the multi-agent coupling channels (shared memory, occupancy) that RAG metrics do not represent; and its diagnoses are validated interventionally (oracle stages; engineered structural faults) rather than by correlation with human ratings. Process mining and conformance checking compare event logs against stage models (van der Aalst 2016; Carmona et al. 2018); the protocol shares that log-level discipline and adds agent-specific event semantics with intervention-based validation. Process reward models score intermediate reasoning steps to supervise training (Lightman et al. 2024); Q/R/M/C is an evaluation-time instrument for memory pipelines.

Claims

We make three claims.

1. **The protocol.** Operational, log-level definitions of the four stage events for single agents (Definition 1) and communicating collectives (Definition 2), with identifiability under a stated conditional stage-Markov assumption, plus a stress test of that assumption.

2. **Stage-level diagnoses.** Single-agent: oracle-stage interventions and log inspection attribute failures to memory accumulation rather than to ranking. Multi-agent: a registered slope-level claim (Empirical Claim 1 below) is confirmed on a 35,640-run sweep, with an interior time-to-success optimum at K=8 and a provenance-level mechanism.

3. **External validity.** A two-layer contract (invariant event roles plus a system-specific adapter) instruments OpenAI-SDK, LangGraph, and AutoGen agents with one shared tool-trace adapter and no per-framework changes; engineered structural failures localise identically on all stacks, and behavioural differences between stacks are attributed stage-by-stage.

Empirical Claim 1 (bottleneck shift, slope-level). As the peer-broadcast cadence K (§) decreases (faster sharing), $P(M^\star|R^\star)$ rises, because fresher peer evidence enables reliable commitment, and $P(C^\star|M^\star)$ falls, because peers compete for the same scarce targets. Both are claims about conditional rates, not about their product.

Table 1: Portability of the contract across system classes: the same four events, defined once, instrument each class without modification and localize a different failure in each. The subsections of § instantiate the rows in order.

System class	Failure mode identified by the four events
Symbolic single agent	ranking 96.9% vs. empty candidates 91%: an *accumulation* limit
Symbolic collective	M↔C bottleneck shift; t_{succ} optimum K=8; provenance mechanism
Learned policies	competitive success, collapsed stage observability
LLM agents	$1.00 \to 0.00$ collapse = pure C-budget; sharing lowers success $2/3 \to 1/3$
Third-party stacks × 2 models	structural failures localize identically; M_1 differs by model and is framework-dependent

Scope and non-claims. (i) The contribution is the operationalisation: log-level event definitions under which the four rates are identifiable on any system exposing a query/lock interface; the chain-rule arithmetic itself is standard. (ii) Empirical Claim 1 is a slope-level empirical observation, not a theorem; both slopes were confirmed with cluster-robust confidence intervals and are robust to the event-definition thresholds (θ, ϵ; supplement). (iii) No claim is made about the conditional product Φ (monotone in our data); the interior optimum concerns conditional time-to-success, and a censoring-aware contrast under which it does not persist is reported. (iv) Primary substrates are grid worlds, which admit oracle interventions and exhaustive sweeps; the VMAS, LLM, and third-party studies indicate that the event definitions do not depend on grid structure. (v) Multi-agent place identity assumes a shared coordinate frame; a companion manuscript removes the assumption without changing the event definitions. (vi) $R^\star$ and $M^\star$ presuppose an entity correspondence between memory content and world referents, supplied here by instrumented environments; interchangeable agents in a single environment inherit it from the shared substrate, and for heterogeneous open-world agents it remains open (the protocol then reduces to $Q^\star$, $C^\star$). Extended discussion of each point is in the supplement.

Method and Materials

Event definitions and estimators

An episode issues a semantic query (required/preferred/penalty tags); memory returns candidates; the planner locks one; the controller executes. The four events anchor at logged triggers on the episode's query–lock chain and are *nested by construction*: $Q^\star$ is recorded on intent emission; $R^\star := Q^\star \wedge$ (the returned candidate set contains a task-satisfying candidate); $M^\star := R^\star \wedge$ (the planner locks a true target drawn from that candidate set); $C^\star := M^\star \wedge$ (the controller reaches the locked target). An episode may run several query–retrieve–lock attempts; a link is recorded if some attempt realizes it together with its predecessors, so $C^\star \subseteq M^\star \subseteq R^\star \subseteq Q^\star$ holds at the episode level by definition. $C^\star$ is *semantic-path completion* — completion through the retrieval path — and is distinct from raw task success, which can also occur off-path.

Two regimes of use. We keep two kinds of quantities strictly separate. *Link rates* are the conditionals $q = P(Q^\star)$, $r = P(R^\star \mid Q^\star)$, $m = P(M^\star \mid R^\star)$, $c = P(C^\star \mid M^\star)$, estimated by conditional frequency over episodes; under the nesting above, $P(C^\star) = q\,r\,m\,c$ is an exact identity — a chain rule that needs no further assumption. *Marginal event frequencies*, written $f(E)$, are used as four separate behavioural diagnostics on systems whose logs expose no lock chain (§); there an event can occur without its predecessor, the events are not nested, and the marginals are never substituted into the product.

Definition 1 (single-agent estimability, informal). With the nested events above, $P(C^\star) = q\,r\,m\,c$ holds identically and each link rate is consistently estimable from logs. The conditional stage-Markov property (each stage outcome depends on the trace only through its predecessor stage) is not needed for the identity; it is what justifies interpreting each link rate as a stable property of the owning subsystem, and hence predicting the effect of a stage intervention. A simulation stress test quantifies the estimator bias when the assumption is violated: with a hidden confounder, m stays unbiased while c inflates, so diagnoses that rest on the M–C *contrast* degrade gracefully (supplement).

Definition 2 (multi-agent extension, informal). With N agents sharing memory by peer broadcast every K ticks, the conditioning sets enlarge with two coupling channels: the shared memory $\mathcal{M}_{-i}^{(K)}$ and occupancy $\mathcal{O}_{-i}$ (scarce targets are consumed). The same four rates are logged per agent.

Interpretive readings of the stages (an order-theoretic account of empty retrieval via the query–extent Galois connection; an options/semi-MDP account of why optima appear on time-bearing metrics) are given in the supplement.

Benchmarks, systems, and interventions

Single agent. The benchmark comprises four MiniGrid task families — water and rest semantic search, goal-region rejoin in FourRooms, and hazard recovery in LavaGapS7 — at 10 seeds $\times$ 8 episodes, with a BabyAI portability check in the supplement. Oracle-stage interventions replace the output of one stage with ground truth while preserving the interface (oracle-R substitutes a correct retrieval, oracle-M a correct lock, oracle-C direct arrival). A learned behaviour-cloning baseline serves as an observability control.

Multi-agent. A scarce water-search task runs over one symbolic substrate under four sharing architectures: independent private memories, a shared event bus, a central trust-weighted aggregator, and peer-to-peer broadcast at cadence $K \in \{1, \ldots, 64\}$ ($N \in \{3, 5, 8\}$ agents, $T \le N$ targets, three layouts, three hazard densities, 40 seeds; 35,640 runs; an $N{=}12$ scale-up of 8,640 runs in the supplement). For provenance analysis, each $M^\star$ is annotated with the foreign share φ of its supporting evidence mass, computed from the merge's own weights and frozen at lock time; locks with $\varphi \ge 0.8$ are termed *warps*. Learned and LLM counterparts: a CommNet-PPO baseline on the same scenario, and a collective of three LLM agents (llama3.1) with shared symbolic memory on a text substrate, paired with a no-communication baseline on identical layouts.

Third-party stacks. The contract has two layers: an *invariant core* — four event roles (a memory query is issued; returned evidence satisfies the task; the agent commits to a target; the task completes) — and a *system-specific adapter* that maps logged actions to those roles. One tool-trace adapter, shared by an OpenAI-SDK function-calling loop, LangGraph's ReAct agent, and AutoGen, exposes an episodic household memory behind `recall`/`goto`/`take` tools; $Q^\star$ is recorded on querying, $R^\star$ on attribution (a returned record names the item at its true place), $M^\star$ on a commitment (`goto`) to the true place, $C^\star$ on completion within a tool budget. A tool trace exposes no lock chain, so these are the marginal event frequencies $f(\cdot)$ of §: an agent can commit to the true place without a supporting retrieval, the events are not nested, and no factorisation is claimed. What varies across system classes is the adapter, not the event roles; two revisions of this adapter are recorded in the pre-run registration. Four variants engineer one failure each; all predictions were registered before the runs, and the full set of 240 episodes was repeated with a second model (qwen3:4b) under a prediction registered in advance. For the blinded fault-localization study, ten fault types — two to three per stage, structural and behavioural — were injected, and a first-failing-link decision rule was fixed, with all thresholds, before any run; the rule receives only anonymised marginal profiles plus a labelled control cell, and a held-out control enters the blind set to measure false positives.

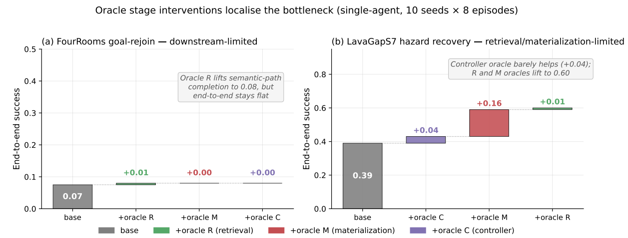

Figure 1: Oracle-stage interventions on the single-agent benchmark lift the stage that the decomposition identifies. Hazard recovery is retrieval-limited (oracle-R: $0.39 \to 0.60$ end-to-end); goal-region rejoin is downstream-limited (oracle-R restores semantic-path completion, end-to-end unchanged).

Results and Discussion

Single agent

End-to-end success hides heterogeneity: success is high on water and rest (0.95, 0.93) and low on rejoin and hazard recovery. The causes differ, and oracle-stage interventions separate them. Replacing retrieval with an oracle lifts hazard-recovery success $0.39 \to 0.60$ (bootstrap 95% CIs $[0.35, 0.43]$ vs. $[0.51, 0.69]$; per-seed paired deltas positive in $9/10$ seeds, exact sign test $p{=}0.004$; retrieval-limited); the same oracle restores semantic-path completion on goal-region rejoin without changing end-to-end success (downstream-limited). The framework's own logs refine the attribution further: when candidates exist, ranking precision is 96.9%, but the candidate set is empty at 91% of decision points — a figure robust to the weighting (0.908 pooled over calls, 0.909 seed-weighted, 0.891 episode-weighted), so it is not an artefact of long failing episodes contributing more decision points. The binding constraint is therefore accumulation: at most decision points there are no candidates to rank. This attribution changed the development priority from ranking to consolidation; success-only evaluation provides no indication of it. Accumulation itself is not one of the four stages: the protocol localises the failure to the R link, and within-link log inspection identifies the cause. We therefore read Q/R/M/C as coarse localisation with a designated drill-down, not as a complete causal ontology of memory failure.

The learned behaviour-cloning baseline reaches competitive success with collapsed stage structure (there is no explicit query/lock interface), so end-to-end optimisation eliminates the events the protocol requires.

Multi-agent

Figure 2 shows the stage profile of every architecture; retrieval saturates, and the differences between architectures are confined to the M and C links, consistent with Empirical Claim 1.

Bottleneck shift. Across peer runs, Spearman of $P(M^\star|R^\star)$ vs. K is -0.53 (cluster-robust 95% CI $[-0.59, -0.47]$ over 81 configuration cells) and of $P(C^\star|M^\star)$ vs. K is $+0.43$ ($[+0.37, +0.50]$). The sharper signature is within-cadence: Pearson between the two links inside a fixed K peaks at -0.61 at $K{=}4$ and decays to noise by $K{=}64$, so the bottleneck migrates between stages within the same configuration as K varies. Scaling to $N{=}12$ sharpens the signature (peak -0.70, migrating to $K{=}8$). On mean time-to-success among successful runs the trade-off produces an interior optimum at $K{=}8$ (7.63 vs. 8.04 independent, cell-weighted): cell-paired contrasts show $K{=}8$ faster than $K{=}16$ (-0.39, 95% CI $[-0.80, -0.08]$) and than independent (-0.40 $[-0.59, -0.24]$), and statistically tied with $K{\in}\{2, 4\}$. That metric conditions on success. Under a censoring-aware alternative that charges failed runs the full step limit, the fast end reverses and independent is marginally best (paired $\Delta(K{=}8 - \text{indep}) = +0.32$ $[+0.05, +0.63]$): the completion decline under fast sharing reduces the success rate by more than the materialization gain increases it. The interior optimum is therefore a statement about conditional speed, not unconditional efficiency; consolidation rate remains a tunable design parameter, and the reservation protocol described below closes the unconditional gap. Oracle interventions confirm the attribution: at fast cadence, oracle-R/M/C progressively remove the excess time, placing it at the M$\leftrightarrow$C transition.

Provenance stratification. The warp share falls from 0.55 at $K{=}1$ to 0.11 at $K{=}8$, while own-evidence locks complete at a flat 0.26–0.34 and warp locks stay below 0.05 throughout. The completion decline under fast sharing is therefore concentrated in the foreign-evidence stratum: cadence controls how much of the collective's commitment relies on peers' evidence, and completions in this stratum are the ones lost to occupancy contention. The stratification is not an artefact of the 0.8 cutoff: recomputed at thresholds $\varphi{\in}\{0.5, \ldots, 0.95\}$, the foreign stratum completes at $\leq$0.03 while the own stratum stays at 0.26–0.50, at every threshold and cadence. The two coupling channels also separate interventionally with arms already in the design: the independent arm removes peer evidence while leaving target contention in place, and a decentralized reservation-and-rollback protocol (companion manuscript) keeps peer evidence while managing contention — under it, unconditional success at fast cadence exceeds the $K{=}8$ baseline (0.614–0.620 vs. 0.583), identifying occupancy contention rather than evidence quality as the cost channel of fast sharing. The same stratification curve reproduces on a continuous VMAS bridge ($0.72 \to 0.00$) and on an LLM text substrate ($0.42 \to 0.08$).

Table 2: Marginal Q/R/M/C event frequencies on third-party stacks. $f(E)$ denotes the fraction of 20 episodes in which event E occurred — a descriptor of behaviour, *not* a link of the chain-rule product: $Q^\star$ — the agent queried memory at least once; $R^\star$ — a returned record named the item at its true place; $M^\star$ — the agent committed (`goto`) to the true place, with or without retrieval support; $C^\star$ — `take` succeeded within the tool budget; M_1 — the first commitment targeted the true place. The events are not nested and no product is claimed: under the consolidation gap, budgeted blind search over the four places sometimes commits to and reaches the true place, hence $M^\star, C^\star > 0$ at $R^\star{=}0$.

Stack	Variant	$f(Q^\star)$	$f(R^\star)$	$f(M^\star)$	$f(C^\star)$	$f(M_1)$
OpenAI SDK	control	1.00	1.00	.85	.75	.25
	consol. gap	1.00	**.00**	.50	.40	.25
	stale ambig.	1.00	1.00	.90	.30	.25
	tight budget	1.00	1.00	.25	**.00**	.25
Lang-Graph	control	1.00	1.00	.70	.50	.50
	consol. gap	1.00	**.00**	.20	.15	.10
	stale ambig.	1.00	1.00	.65	.60	.50
	tight budget	1.00	1.00	.10	**.00**	.10
Auto-Gen	control	1.00	1.00	.85	.55	.45
	consol. gap	1.00	**.00**	.30	.20	.15
	stale ambig.	1.00	1.00	.50	.45	.35
	tight budget	1.00	1.00	.15	**.00**	.15

Learned and LLM collectives. The CommNet-PPO baseline matches symbolic peer success (0.667 vs. 0.65) with structurally degenerate stages: its real-valued channel is never informatively empty, so R and M collapse and the stage events are not individually observable despite competitive success. The LLM collective reproduces the same pattern on a language substrate: paired against the no-communication baseline on identical layouts, raw sharing lowers success $2/3 \to 1/3$ (occupancy contention), and the reservation-and-rollback protocol recovers part of it ($0.33 \to 0.42$) with rollback latency $\approx K$. On a step-budget sweep with two local models, a success collapse that is opaque to end-to-end evaluation ($1.00 \to 0.00$) resolves to pure C-stage failure ($Q^\star{=}R^\star{=}M^\star{=}1, C^\star{=}0$) identically for both models, which identifies the failure as structural rather than model-specific.

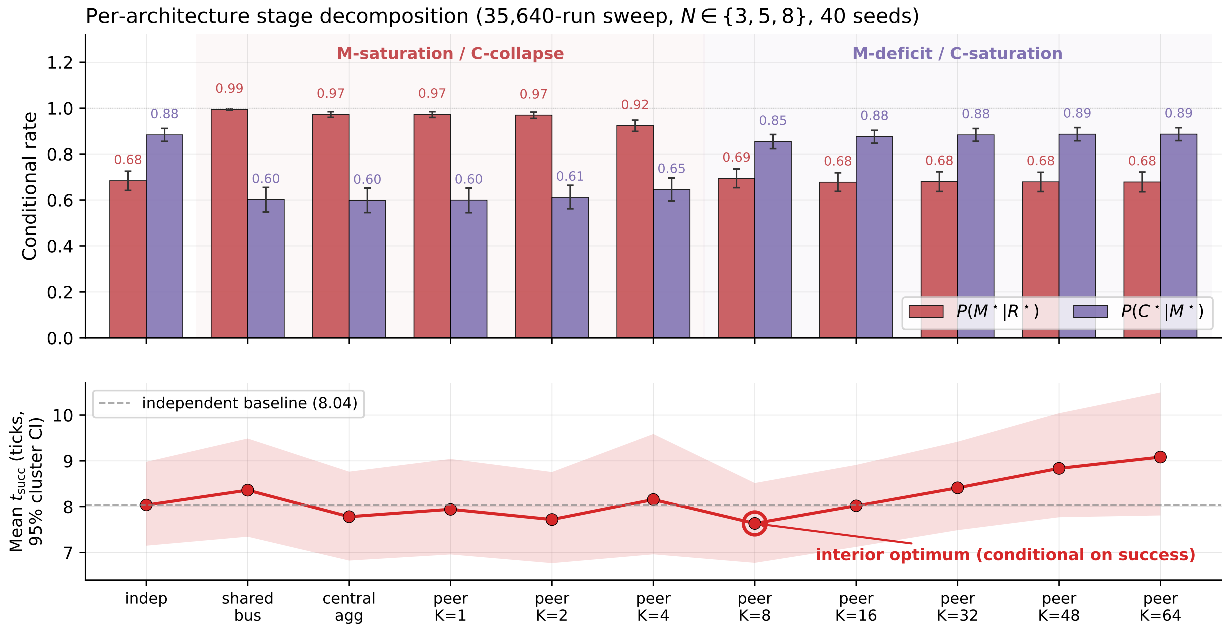

Figure 2: Per-architecture conditional Q/R/M/C rates (top) and mean time-to-success (bottom) on the 35,640-run sweep. Shared bus, central aggregator, and fast peer broadcast ($K \leq 4$) sit in the M-saturation/C-collapse regime; independent and slow peer ($K \geq 16$) in the reverse. Error bars and band: 95% cluster-bootstrap CIs over the 81 design cells. Time-to-success among successful runs has an interior minimum at $K{=}8$ (7.63 vs. 8.04 independent); cell-paired contrasts and a censoring-aware counterpart are reported in the text.

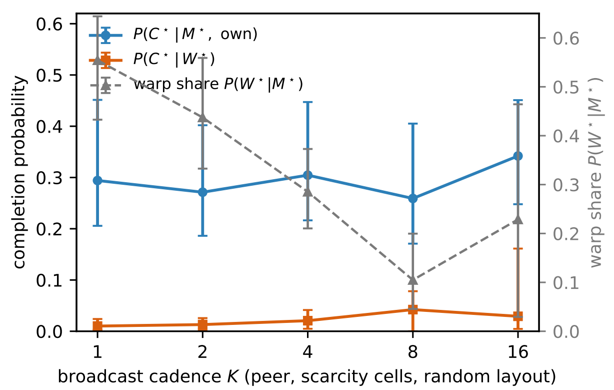

Figure 3: Provenance stratification of $P(C^\star|M^\star)$ vs. cadence K (peer, scarcity cells, episode-cluster bootstrap CIs). The share of locks backed by foreign evidence (warp share, grey) grows as sharing accelerates, while that stratum's completion (orange) stays below 0.05 and the own-evidence stratum (blue) stays flat: the C-collapse of fast sharing is concentrated in the foreign-evidence stratum.

Third-party stacks

Three findings emerge (Table 2, Fig. 4). **(1) Structural localizations are exact and framework-independent.** With no consolidated fact, $f(R^\star){=}0.00$; with the budget below requirement, $f(C^\star){=}0.00$ at $f(Q^\star){=}f(R^\star){=}1$. Both hold in $20/20$ episodes on every stack with no per-framework adaptation. **(2) The clean control separates the stacks.** Success on identical clean episodes spreads 0.50–0.75. The stage profiles attribute this spread to a failure mode common to all three stacks — low first-commitment accuracy ($f(M_1){=}0.25$–0.50: the first `goto` frequently does not use the agent's own retrieved evidence) — combined with framework-specific recovery: $f(M^\star)$ recovers to 0.70–0.85 through re-querying, and the fraction reaching completion differs across stacks (C/M from 0.88 to 0.65). **(3) The cost of the stale distractor is framework-specific.** An outdated record attracts the first commitment in 0.20–0.40 of ambiguous episodes, and the completion cost varies (OpenAI SDK: $0.75 \rightarrow 0.30$; the other stacks recover from the detour). Several registered behavioural predictions assumed a clean control and failed for that reason: the stacks themselves fail a clean memory task in 25–50% of episodes (finding 2). These are reported as registered failures together with the mechanism.

Model robustness. Repeating all 240 episodes with qwen3:4b matches the registered prediction: both structural localizations transfer unchanged ($f(R^\star){=}0.00$ under the consolidation gap and $f(C^\star){=}0.00$ under the tight budget, $20/20$ on every stack). The behavioural profile, by contrast, is a model property that the stage events make observable. qwen3:4b completes every clean episode ($f(C^\star){=}1.00$ on all three stacks) with first-commitment accuracy $f(M_1){=}0.90$–0.95, vs. 0.25–0.50 for llama3.1, and its first commitment follows the stale record in only 0.05–0.30 of ambiguous

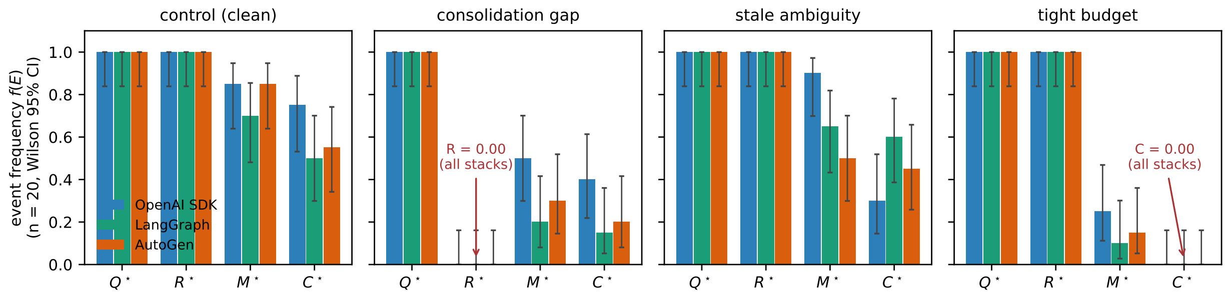

Figure 4: Stage profiles of three third-party stacks on four engineered variants (llama3.1, n=20/cell). Q and R saturate identically; the structural failures give exact zero columns on every stack; the inter-framework spread is confined to M and C.

episodes. The exception is AutoGen, under which qwen's $f(M_1)$ falls to 0.40; the decomposition thus separates the model's contribution from the effect of the framework's prompting loop. The full second-model table is in the supplement.

Blinded fault localization. Stage-level accuracy is $26/30$ scored cells for llama3.1 and $29/33$ for qwen3:4b; the five structural faults localize $30/30$ across both models and all three stacks, and held-out controls read "none" in $5/6$ cells (one false positive at n=20). The misclassifications follow an interpretable pattern. M-stage faults are masked by llama3.1's low baseline $f(M_1)$ (0.25–0.50) and become detectable on qwen3:4b's high baseline, as registered before the runs: the unmarked duplicate distractor lowers $f(M_1)$ by $0.90/0.80/0.20$ across the three stacks. The *marked* stale distractor changes llama3.1's behaviour but not qwen3:4b's, which takes the staleness marker into account; the rule accordingly reports no failure for qwen3:4b. Registration, decision rule, and per-fault outcomes are in the supplement.

Limitations

Primary substrates are grid worlds; VMAS is a directional continuous check; the LLM studies use small local models at modest n; place identity across agents assumes a shared coordinate frame (removed constructively in a companion manuscript); Empirical Claim 1 remains slope-level; a derivation under a generative model would make it a theorem. The four stages are one resolution level: sub-stage causes (accumulation within R; ranking, grounding, and staleness handling within M) require within-link log inspection, which the protocol scopes but does not automate. Full tables, proofs of the estimability arguments, assumption stress tests, effect-size methodology, scale-up, and all registered-prediction files are in the supplement.

Conclusion

Q/R/M/C replaces a scalar success measure with a stage-level diagnosis. In our studies the protocol identified a consolidation limit in a single agent, an occupancy-governed sharing-rate trade-off in a collective, and low first-commitment accuracy in off-the-shelf agent stacks; two of these diagnoses led to verified modifications (consolidation-directed development; the reservation protocol). The protocol requires a logging contract rather than architectural modification, and the resulting rates are directly comparable across systems that expose a query/lock interface; for opaque learned policies the protocol measures stage observability itself.

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Reproducibility Checklist

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```
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1. General Paper Structure

1.1. Includes a conceptual outline and/or pseudocode description of AI methods introduced (yes/partial/no/NA) partial

1.2. Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results (yes/no) yes

1.3. Provides well-marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper (yes/no) yes

2. Theoretical Contributions

2.1. Does this paper make theoretical contributions? (yes/no) yes

If yes, please address the following points:

2.2. All assumptions and restrictions are stated clearly and formally (yes/partial/no) yes

2.3. All novel claims are stated formally (e.g., in theorem statements) (yes/partial/no) partial

2.4. Proofs of all novel claims are included (yes/partial/no) partial

2.5. Proof sketches or intuitions are given for complex and/or novel results (yes/partial/no) yes

2.6. Appropriate citations to theoretical tools used are given (yes/partial/no) yes

2.7. All theoretical claims are demonstrated empirically to hold (yes/partial/no/NA) yes

2.8. All experimental code used to eliminate or disprove claims is included (yes/no/NA) yes

3. Dataset Usage

3.1. Does this paper rely on one or more datasets? (yes/no) yes

If yes, please address the following points:

3.2. A motivation is given for why the experiments are conducted on the selected datasets (yes/partial/no/NA) yes

3.3. All novel datasets introduced in this paper are included in a data appendix (yes/partial/no/NA) NA

3.4. All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no/NA) NA

3.5. All datasets drawn from the existing literature (potentially including authors' own previously published work) are accompanied by appropriate citations (yes/no/NA) yes

3.6. All datasets drawn from the existing literature (potentially including authors' own previously published work) are publicly available (yes/partial/no/NA) yes

3.7. All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisficing (yes/partial/no/NA) NA

4. Computational Experiments

4.1. Does this paper include computational experiments? (yes/no) yes

If yes, please address the following points:

4.2. This paper states the number and range of values tried per (hyper-) parameter during development of the paper, along with the criterion used for selecting the final parameter setting (yes/partial/no/NA) partial

4.3. Any code required for pre-processing data is included in the appendix (yes/partial/no) yes

4.4. All source code required for conducting and analyzing the experiments is included in a code appendix (yes/partial/no) yes

4.5. All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no) yes

4.6. All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from (yes/partial/no) partial

4.7. If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results (yes/partial/no/NA) yes

4.8. This paper specifies the computing infrastructure used for running experiments (hardware and software), including GPU/CPU models; amount of memory; operating system; names and versions of relevant software libraries and frameworks (yes/partial/no) yes

4.9. This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics (yes/partial/no) yes

4.10. This paper states the number of algorithm runs used to compute each reported result (yes/no) yes

4.11. Analysis of experiments goes beyond single-dimensional summaries of performance (e.g., average; median) to include measures of variation, confidence, or other distributional information (yes/no) yes

4.12. The significance of any improvement or decrease in performance is judged using appropriate statistical tests (e.g., Wilcoxon signed-rank) (yes/partial/no) yes

4.13. This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments (yes/partial/no/NA) yes